
THE INDICATORS, FROM THE FORMULA UP

Jonas Löffler

loefflerjonas6@gmail.com

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Abstract

Most traders meet technical indicators as coloured lines with rules attached and never see the arithmetic underneath. This paper does the opposite: it derives the standard indicators from their formulas, so that each one stops being a black box and becomes a transparent transform of price. We work through the two families a chartist actually uses — *smoothers* (simple, weighted, and exponential moving averages, and the MACD built from them) and *oscillators* (RSI, the stochastic, Bollinger Bands, ATR, and the ADX/DMI system) — writing out each definition, the single thing it measures, and the specific way it misleads. The unifying result is deflationary and important: every one of these indicators is a deterministic function of the same price and volume series, so none of them adds information that is not already in the chart; they only *reshape* it to make one feature — trend, momentum, volatility, or dispersion — easier to read. Understanding the formula is what lets a trader see when an indicator is saying something real and when it is merely re-describing the price with a lag.

Keywords — Moving average, EMA, MACD, RSI, stochastic oscillator, Bollinger Bands, ATR, ADX, directional movement, lag, smoothing, volatility, indicator overfitting

1 Introduction

1.1 Motivation and thesis

An indicator is a promise: "compute this number from recent prices, and it will tell you something the raw chart hid." Sometimes the promise is kept — a volatility measure genuinely summarises dispersion you would struggle to eyeball — and sometimes it is empty, a lagged copy of price dressed up as a signal. The only way to tell which is which is to know the formula. The thesis of this paper is that every popular indicator is a *deterministic transform* of the price (and sometimes volume) series, so it can add no information that was not already present; its value is entirely in whether that transform makes a genuine feature — trend, momentum, volatility — easier and more honest to read.

1.2 Scope, prerequisites, and intended reader

You need to know what a price series is and be comfortable with an average, a percentage, and a standard deviation; nothing else. Every symbol is defined where it first appears. This paper is a *reference*: it states each indicator precisely enough that you could implement it yourself and know exactly what your charting package is drawing. It is the companion to the conceptual technical-analysis paper, which argues *whether* rules built on these indicators survive testing; here we only build the instruments, honestly labelled.

1.3 Notation and reproducibility

Let P_t be the closing price on day t , and where an indicator needs them, H_t , L_t the day's high and low. All definitions below are the standard public forms; where a smoothing convention has a common variant (notably Wilder's smoothing versus a textbook exponential average), it is named. Every indicator here is a few lines of arithmetic and can be reproduced from free daily data with no special library. *This is a research and educational document, not investment advice.*

1.4 Organization of this paper

Section 2 draws the map: two families, smoothers and oscillators, and what separates them. Section 3 builds the moving averages (simple, weighted, exponential) and the lag they all pay. Section 4 assembles the MACD from two of them. Sections 5 and 6 build the momentum oscillators RSI and the stochastic. Section 7 builds Bollinger Bands from a moving average and a standard deviation. Section 8 builds ATR as a volatility measure, and Section 9 the ADX/DMI trend-strength system. Section 10 states the deflationary result — no indicator adds information

— and its consequences for overfitting. Section 11 collects misconceptions, Section 12 is a glossary, and the references close the paper.

2 Two families of indicator

Almost every indicator on a standard platform belongs to one of two families, and knowing which tells you in advance what it can and cannot do. *Smoothers* average price over a window to expose the slow component — the trend — at the cost of *lag*: because they look backward, they turn after the price does. Moving averages and the MACD are smoothers. *Oscillators* normalise recent price action onto a bounded or centred scale to expose a fast component — momentum, or distance from a local mean — and are prone to the opposite error: they flag "overbought" or "oversold" states that persist for a long time in a trend. RSI, the stochastic, and the %B reading of Bollinger Bands are oscillators. A third, smaller group measures *dispersion* rather than direction — ATR and the band width — and a fourth measures *trend strength without direction* — the ADX. The families are not decoration; a smoother used as a timing trigger and an oscillator used to fight a trend fail for reasons written into their formulas.

3 Moving averages and the price of smoothing

A *simple moving average* (SMA) of length n is the plain mean of the last n closes:

$$\mathrm{SMA}_n(t) = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i}.$$

It smooths the series but reacts slowly and weights a price from $n-1$ days ago exactly as heavily as today's. A *weighted moving average* (WMA) fixes the second complaint by giving linearly larger weight to recent prices, and an *exponential moving average* (EMA) fixes it recursively and cheaply:

$$\mathrm{EMA}_t = \alpha P_t + (1-\alpha) \mathrm{EMA}_{t-1}, \quad \alpha = \frac{2}{n+1}.$$

Here α is the *smoothing factor*: a larger α (shorter n) tracks price closely and stays jumpy; a smaller α (longer n) is smoother and slower. One smoothing convention is worth naming because several later indicators use it: *Wilder's smoothing* is an EMA with $\alpha = 1/n$ rather than $2/(n+1)$, which is why a "14-period" Wilder average is slower than a "14-period" textbook EMA. Every moving average shares one unavoidable property — *lag*. Because it is an average of past prices, it necessarily turns after the price turns, and the smoother you make it, the later it turns. That lag is not a bug to be tuned away; it is the price you pay for suppressing noise, and it is why moving-average crossovers whipsaw in ranges and shine only when a trend runs far enough to outpay the late entry and late exit.

4 MACD: two moving averages in a trench coat

The *moving average convergence/divergence* is not a new idea but a packaging of Section 3. It is the difference of two EMAs, conventionally 12- and 26-period, with a third EMA of that difference as a *signal line*:

$$\begin{aligned} \mathrm{MACD}_t &= \mathrm{EMA}_{12}(P)_t - \mathrm{EMA}_{26}(P)_t, \quad \mathrm{signal}_t = \mathrm{EMA}_9(\mathrm{MACD})_t, \end{aligned}$$

and the *histogram* is $\mathrm{MACD}_t - \mathrm{signal}_t$. Because EMA_{12} reacts faster than EMA_{26} , their difference is positive when the fast average is above the slow one — a short-term uptrend relative to a longer one — and its sign changes are just crossovers of the two underlying averages, seen through one number. The histogram, being the gap between the MACD and its own smoother, is essentially a measure of how fast the trend relationship is *accelerating*. Knowing this deflates most MACD folklore: a "bullish MACD cross" is exactly a fast-EMA-over-slow-EMA cross, carries the same lag as any moving-average system, and adds no information beyond the two averages it is built from.

5 RSI: bounded momentum

The *Relative Strength Index* maps recent up-moves against recent down-moves onto a 0–100 scale. Over a window (Wilder's original is $n = 14$), let the average gain be the Wilder-smoothed mean of the up-closes and the average loss the Wilder-smoothed mean of the down-closes (as a positive number). Then

$$\begin{aligned} \mathrm{RS} &= \frac{\text{average gain}}{\text{average loss}}, \quad \mathrm{RSI} = 100 - \frac{100}{1 + \mathrm{RS}}. \end{aligned}$$

When gains dominate, RS is large and RSI approaches 100; when losses dominate, RSI approaches 0; when they balance, RSI sits at 50. The conventional 70/30 thresholds label "overbought" and "oversold" states. The formula makes the standard warning precise: RSI is a bounded transform of the *ratio* of recent gains to losses, so in a persistent trend the ratio stays lopsided and RSI can pin near an extreme for weeks while price keeps going. Reading a single RSI reading as a reversal signal is therefore reading a momentum measure as if it were a mean-reversion one — safe only after you have separately established a mean-reverting regime.

6 The stochastic oscillator: position within the range

Where RSI works from gains and losses, the *stochastic oscillator* works from *where price sits inside its recent range*. Over a look-back of n periods,

$$\%K_t = 100 \cdot \frac{P_t - \min_{i < n} L_{t-i}}{\max_{i < n} H_{t-i} - \min_{i < n} L_{t-i}}, \quad \%D_t = \text{SMA}_3(\%K)_t$$

$\%K$ is 100 when the close is at the top of its n -period range and 0 when it is at the bottom; $\%D$ is a 3-period smoothing of $\%K$ used as a signal line. The intuition is that closes near the top of the range signal strength and near the bottom signal weakness, and that turns in $\%K$ relative to $\%D$ hint at momentum shifts. The same caveat as RSI applies and for the same structural reason: in a strong trend the close sits near the top (or bottom) of its range day after day, so the stochastic saturates and its "overbought" reading is not a warning but a description of a healthy trend.

7 Bollinger Bands: a moving average with a volatility envelope

Bollinger Bands wrap a moving average in a band whose width scales with recent volatility. With a 20-period SMA and the sample standard deviation σ_{20} of the last 20 closes,

$$\text{middle} = \text{SMA}_{20}(t), \quad \text{upper/lower} = \text{SMA}_{20}(t) \pm k \sigma_{20}(t)$$

with $k = 2$ conventionally. Because the band is $\pm k$ standard deviations, it automatically widens when the market is volatile and narrows when it is calm — the much-discussed "squeeze" is simply σ_{20} falling. The bands answer a precise question: how far is price from its recent mean, measured in units of its own recent volatility? A close outside a band means an unusually large move *relative to recent dispersion*, nothing more — it is a statement about volatility-scaled distance, not a reversal signal, and in a trend price can "walk the band" for a long time.

8 ATR: measuring how much it moves

The *Average True Range* strips out direction entirely and measures only how much an instrument moves per day, correctly accounting for gaps. The *true range* on day t is the largest of the day's own range and the two gap-adjusted ranges to the prior close:

$$\text{TR}_t = \max(|H_t - L_t|, |H_t - P_{t-1}|, |L_t - P_{t-1}|)$$

and the ATR is a Wilder-smoothed average of TR , conventionally over 14 periods. ATR is a volatility measure in price units, and its honesty is exactly that it says nothing about direction. Its genuine uses are practical rather than predictive: sizing a position so that a fixed dollar risk corresponds to a fixed multiple of ATR, or setting a stop a volatility-scaled distance away so that the same stop is wide in turbulent markets and tight in calm ones — the same volatility-normalisation logic that runs through the risk and volatility papers.

9 ADX and DMI: trend strength without direction

The *Directional Movement* system separates the question "is there a trend?" from "which way?". From the day-to-day changes in the high and low it forms *directional movement* up and down, smooths each (Wilder) and divides by ATR to get the positive and negative *directional indicators* $+\mathrm{DI}$ and $-\mathrm{DI}$. Their normalised gap is the *directional index*,

$$\mathrm{DX} = 100 \cdot \frac{|+\mathrm{DI} - -\mathrm{DI}|}{+\mathrm{DI} + -\mathrm{DI}},$$

and the *ADX* is a Wilder-smoothed average of DX. The two DI lines carry direction — $+\mathrm{DI}$ above $-\mathrm{DI}$ means up-moves dominate — while the ADX carries only *strength*: a high ADX means a strong trend of either sign, a low ADX means a range. This separation is the whole point and the one genuinely distinct idea in the indicator zoo: ADX is the tool that answers the prior question the moving-average and oscillator families both quietly assume — whether the market is trending at all, which decides which family should be trusted right now.

10 The deflationary result: no indicator adds information

Step back and the common structure is unmistakable. Every indicator above is a fixed, deterministic function of P_t (and for a few, H_t , L_t , and volume). A function of the data cannot contain information the data did not — it can only *reshape* what is already there to make one feature legible. That is the honest description of what an indicator does and the boundary of what it can do: a moving average makes trend legible and lags; an oscillator makes short-horizon distance-from-mean legible and saturates in trends; ATR makes volatility legible and is silent on direction; ADX makes trend *strength* legible. None of them is a source of new predictive content, and stacking five of them does not create any — it mostly creates *overfitting*, because each indicator adds parameters (lengths, thresholds) to tune until the past looks explained. The discipline the rest of this library insists on therefore applies with full force: an indicator-based rule is worth exactly as much as it proves out-of-sample, after costs, deflated for the number of indicator-and-parameter combinations you tried — and no more.

11 Common misconceptions

- **"More indicators mean more information."** All indicators are transforms of the same price and volume; adding them adds parameters to overfit, not information (Section 10).
- **"The MACD cross is its own signal."** A MACD sign change is exactly a fast-EMA-over-slow-EMA crossover, with the same lag as any moving average (Section 4).

- **"RSI over 70 means reverse."** RSI is bounded momentum; in a trend it pins near an extreme for weeks (Section 5).
- **"A close outside the Bollinger Band is a reversal."** It is an unusually large move relative to recent volatility; in a trend, price walks the band (Section 7).
- **"ATR tells me where price is going."** ATR measures how much, never which way; its uses are sizing and stops (Section 8).
- **"A high ADX means go long."** ADX measures trend *strength* of either sign; direction lives in the $+\mathrm{DI}/-\mathrm{DI}$ lines (Section 9).

12 Glossary

- **ADX (Average Directional Index)** — a smoothed measure of trend strength, direction-agnostic.
- **ATR (Average True Range)** — a smoothed average of the true range; a volatility measure in price units.
- **Bollinger Bands** — a moving average with an envelope at $\pm k$ standard deviations of recent price.
- **DI ($+\mathrm{DI}$, $-\mathrm{DI}$)** — directional indicators measuring the dominance of up- versus down-moves.
- **EMA (exponential moving average)** — a recursive weighted average with smoothing factor $\alpha = 2/(n+1)$.
- **Lag** — the delay with which any backward-looking average turns after price turns.
- **MACD** — the difference of a fast and a slow EMA, with a signal-line EMA of that difference.
- **RSI (Relative Strength Index)** — a 0–100 transform of the ratio of recent average gain to average loss.
- **SMA (simple moving average)** — the unweighted mean of the last n prices.
- **Stochastic oscillator** — $\%K$, the close's position within its recent high–low range, and $\%D$, its smoothing.
- **Wilder's smoothing** — an exponential average with $\alpha = 1/n$, used in RSI, ATR, and ADX.

References

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All references above are publicly accessible: widely held published books and a peer-reviewed survey article. No proprietary, course, or trading-academy material is cited or used anywhere in this paper; every formula is a standard public definition.

Evidence basis

Claim	Basis
Each indicator is a deterministic transform of price/volume and adds no new information	Own reasoning from the stated formulas; reproducible from free daily data
The MACD sign change is identical to a fast/slow EMA crossover	Own derivation from the definition (Section 4)
RSI and the stochastic saturate in trends because they are bounded transforms of recent momentum	Own derivation (Sections 5–6)
Bollinger Bands widen/narrow with recent volatility because width is $\pm k \cdot \sigma$	Own derivation (Section 7)
RSI, ATR, and ADX use Wilder's $\alpha = 1/n$ smoothing	Literature — (Wilder 1978)
Indicator-based rules must be judged out-of-sample, after costs, deflated for parameter search	Practitioner consensus; method in the backtesting and overfitting-control papers